

APM Project Management Qualification

Learner Study Pack



Resource management

Introduction

This section of the Learner Study Pack will cover the Resource Management competence of the APM Project Management Qualification.

This section will enable you to gain an understanding of identifying and scheduling the resources required to implement a project in an optimal way.

Learning objective and outcomes

Learning objective:

21. Understand resource management as the ability to identify and schedule the required internal and external resources.

By the end of this activity, you should be able to:

21a) Know how to determine the resources required and their availability to deliver activities within a project.

21b) Understand how an organisational breakdown structure is used to create a responsibility assignment matrix (RACI).

21c) Understand how resources are categorised and allocated to both linear and iterative life cycle schedules.

21d) Know the differences between resource smoothing and resource levelling

Study time

This section of the Learner Study Pack will take approximately 1 - 2 hours to complete.

Recommended resources

In addition to your Learner Study Pack, use the below references for more information on Assurance:

- APM Body of Knowledge Seventh edition, 4.2.7 (Resource optimisation)
- APM Learning Module, Resource management and capacity planning. Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
- APM Learning Module, Information sheet: Resource smoothing and levelling. Accessed at: <https://www.apm.org.uk/resources/apm-learning/>
- APM Resources, what is resource optimisation? Accessed at: <https://www.apm.org.uk/resources/what-is-project-management/what-is-resource-optimisation/>

Overview

Resource management focuses on the effective allocation, utilisation and monitoring of resources throughout the project life cycle ensuring that projects are executed efficiently, on schedule, and within budget.

Determining Resources Required

There are several considerations that need to be made when establishing the resource requirements for a project, including:

- Life cycle approach (Refer to Life cycle competency).
- Activities and tasks to be completed.
- Types of resources needed to support activities.
- Skill sets and capabilities needed from resources.
- An understanding of what the critical tasks for resources are.
- Timeframe resources are needed for.
- Needs of the project versus needs of Business as Usual (BAU).
- Any external resources needed.

Resources may also drive the pace of a project by placing constraints on what can be done and when, as well as constrain project logistics through the space needed to operate and store items.

Project Managers work with Resource Managers to secure resources at the required times for the schedule. Resource Managers may provide line management to the resources in BAU and therefore may be allocated to the project.

If the resource has come from a supplier an external resource manager may be allocated to manage relations

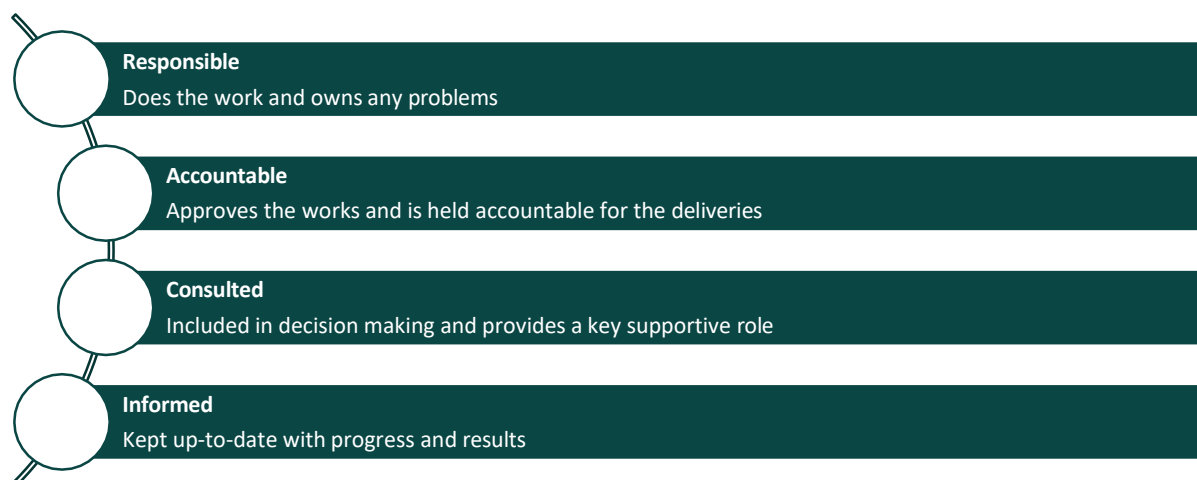
Organisational Breakdown Structure (OBS) and Responsibility Assignment Matrix (RACI)

An organisational breakdown structure (OBS) details the organisation for the project and the resources available. Tasks can then be allocated to these resources defined in the OBS, helping to clarify roles for people. An OBS also makes initial escalation routes clear, which helps with any resource problems that may need escalating. The Project Manager will create a responsibility assignment matrix (RAM) for the project. A common coding structure applied to a RAM is called the RACI matrix:

For each task, the Project Manager will decide if the team member is:

- **Responsible**
- **Accountable**
- **Consulted,**
- **Informed**

It is best practice for one person to be responsible and one person to be accountable.



The RAM or RACI is a communication device to ensure that the people involved are aware of the work they have to do and their position in the project organisation.

Resource managers (these may be Line Managers or BAU Managers) should also consult the RAM/RACI to understand and agree the work that has been allocated to their people.

In summary, the RAM/RACI provides an understanding of the tasks or deliverables, the specific responsibilities defined within the project procedures, and the level of accountability or contribution expected from named roles or individuals within the project.

Categorising and allocating resources

Resources are the labour and non-labour items needed to deliver the project objectives successfully to the agreed acceptance criteria.

Categorised into four types:

1. human
2. materials
3. plant or equipment
4. facilities

When looking to allocate resources the project manager must consider:

- The allocation of resource type.
- How many resources needed for each type.
- Availability to support the schedule.
- Cost.

When using a linear life cycle, a planning assumption made when creating the schedule is all resources will be available when required.

If this is not the case, resource management techniques can be applied to try and fit the work in with the resources available as much as possible maintaining the schedule and quality.

For iterative life cycles, requirements are prioritised and delivered within the timebox using the pre-allocated resources available, varying scope and quality as required.

However, if all of the scope can't be achieved to the agreed quality, an extension of time may be needed.

Resource Smoothing vs. Resource Levelling

Resource management techniques, such as levelling and smoothing, attempt to fit the scope of work into the resource limit, while maintaining the schedule and quality objectives.

Resource Levelling:

- Used where a fixed amount of resource is available, and the end of date of the project can be delayed. It answers the question, ***‘With the resources available, when will the work be finished?’***
- Levelling can be achieved through redefining scope and/or specifications, increasing task durations, increasing resources earlier on to bring activities forward, or moving non-critical activities to a different time.

Resource Smoothing:

- Used when time is more important than cost, answering the question, ‘What resources do I need to deliver the work within the fixed timescale?’ It looks to protect the end date of the project.
- Smoothing can be achieved through reducing task durations by adding more resources, avoiding peaks and troughs of resource demand, or changing the order of activities where the logic used originally was optional to run activities in parallel rather than in sequence.

The overall aim of the techniques is to optimise resources through the flow of work between them, thus preventing the need to change the planned end date of the project.

However, if resources are finite (being either the people or equipment needed to complete the tasks) then there may be no option but to extend project timelines.

Scenario One:

Robin, the Finance Manager, has been reviewing the schedule for the renovation project as they want to make sure that there are no delays in the required resources arriving onsite to hinder the progress of the build.

They have an avid interest in market trends and like to keep an eye on both the local and national economy. They've read that there are several large national civil engineering projects that are about to begin, including several motorway improvements across the country.

Projects of that size require a considerable amount of concrete and therefore this has created a sudden surge in demand. This surge is creating national shortages, and price increases.

This concern has given rise to apprehension about the project overall and how resources have been allocated and managed. Robin is now worried about what happens if the project runs out of other supplies, like steel and bricks.

What needs to be considered when allocating resources for the project?

What tools and techniques could you take Robin through to reassure them that resource management processes have been followed?

Scenario Two:

The 'Snappy Appy' renovation project involves multiple teams and stakeholders, including engineers, contractors, local authorities, and environmental consultants.

Various tasks and decisions must be made and completed throughout the project lifecycle and this requires clear organisational structure and accountability.

How would the utilisation of an Organisational Breakdown Structure (OBS) and a RACI matrix clarify roles, responsibilities, and decision-making authority among the diverse teams and stakeholders involved in the project?

Scenario Three:

Construction work onsite seems to be progressing well, despite the concrete shortage, but some of the subcontractors are facing scheduling conflicts caused by the overlapping of resource demands.

This has the potential to push out the end date of the renovation project and Ayo, the Founder, has been very clear that they do not want this to happen, regardless of the impact on cost.

As the Project Manager, what decisions could you make to ensure resource optimisation and that the deadline is still met?

Learning review questions

Before you attempt the exam practice questions in the next section, we recommend you review your learning to quickly see which aspects you don't understand or can't remember as well.

It will also boost your confidence by confirming what you have learnt and your demonstrated ability to apply that knowledge, so that you are ready to practise for the exam.

Here are some questions you may wish to answer to review your learning from the Resource Management section:

- What are some of the different constraints that projects may need to manage from each of the resource types?
- How could resource management techniques be used to influence the schedule for a project?
- In what ways can the organisational breakdown structure be used to support a project?
- What are the different elements to consider when allocating resources?

Exam practice questions

21.1 You are on a project to implement a new HR platform and you are responsible for resource planning.

Which of the following considerations are not critical to take into account?
(1 mark)

- a. Skillset and capability of resources
- b. Budget constraints
- c. The selected lifecycle of the project
- d. Personal preferences of resources

21.2 You are a project manager tasked with producing a responsibility assignment matrix (RACI) for your new project. You decide to set up a workshop with your project team to complete the RACI.

Which of the statements below would support this task?
(1 mark)

- a. The project manager informs the project team at the workshop which tasks are assigned.
- b. The Work Breakdown Structure (WBS) will be used on its own to clarify the roles against each task.
- c. The Organisational Breakdown Structure (OBS) will be used with the Work Breakdown Structure (WBS) to clarify the roles against each task.
- d. The project manager should ensure every project task is documented as part of the RACI.

21.3 Which of the following will a project manager not need to consider when allocating resources to a schedule in an iterative lifecycle?
(1 mark)

- a. What resources will be needed.
- b. Who will be committing resources.
- c. Whether the required resource is on the critical path.
- d. Whether all resources are available for the entirety of the project.

21.4 You are managing a project with a strict completion date that cannot change. Which of the following resource management techniques would be most suitable?
(1 mark)

- a. Resource smoothing
- b. Resource levelling
- c. Project scheduling
- d. Resource allocation

21.5 You are in the planning stages of a software upgrade project and are required to estimate your planned resources.

When planning your people resources, you need to consider that their availability may impact the _____ (a) _____.

Other types of resources you should consider are _____ (b) _____.

(2 marks)

(a)

1. Choice of solution
2. Project schedule
3. Business case
4. Quality of deliverables

(b)

1. Materials, documents and budgets
2. Facilities, constraints and documents
3. Materials, deliverables and equipment
4. Budget, facilities and equipment

21.6 How is the end of date of a project impacted by:

- Resource smoothing?
- Resource levelling?

(2 marks)

21.7 You have been appointed as the project manager for a large-scale construction project to build a new office complex for a multinational corporation. The project involves multiple stakeholders, intricate timelines, and a diverse range of resources. Explain how resource management plays a crucial role in the successful completion of this project.

In your response:

1. Explain three reasons why it is important to identify and schedule the required resources. (3 marks)
2. Explain how using a responsibility assignment matrix (RAM) can support resource management. (2 marks)

21.8 How are resources categorised and allocated to linear life cycle schedules?

(3 marks)

Why would this approach to resource allocation be challenging for iterative life cycle schedules?

(2 marks)

Exam practice answers

Question	Answer / mark scheme
21.1	D
21.2	C
21.3	D
21.4	A
21.5	(a) 2 (b) 4
21.6	1 mark for how resource smoothing impacts the end date of a project: • The end date stays the same / is maintained / is not impacted • The end date is prioritised • The time of the project is not impacted 1 mark for how resource levelling impacts the end date of a project: • The end date may change / is impacted / is not maintained • The end date is not prioritised • The time of the project is impacted
21.7	1 mark for each reason why identifying and scheduling resources is important, including, but not limited to the below – 3 marks available in total: • It ensures all required resources are made available when needed. • It ensures resources are allocated effectively, avoiding bottlenecks and clashes. • It maximises productivity / efficiency. • It supports effective budgeting, avoiding overspend. • It manages risk around any resource gaps/conflict. • It improves stakeholder/team coordination. 1 mark for how a RAM can support resource management, including but not limited to the below – 2 marks available in total: • It can clarify roles and responsibilities. • It can streamline the allocation of resources. • It can support effective task delegation to resources. • It can be a point of reference for the project team.
21.8	1 mark for each valid point, any of the below – 3 marks available in total: • Resources are categorised according to their type (human resources, materials, equipment, financial resources). • Resources are allocated to specific project tasks or work packages / A Work Breakdown Structure can be used to identify work needed and then the resources can be allocated. • Involves upfront planning and estimation to determine resources needed. 1 mark for each valid point, any of the below – 2 marks available in total: • As iterative life cycles have more flexibility / less structured, resource requirements can be more uncertain / fluctuating / changing. • As iterative life cycles have fixed / pre-allocated resources, it can be difficult to adapt to the project requirements / needs. • As iterative life cycles have fixed / pre-allocated resources, it could lead to under/over-utilisation of resources depending upon the project requirements / needs.